Part 2

ENERGY STORAGE SOLUTION Enters Iron Age

After exploring the working principles and general design features of iron-air batteries in Part 1, in this ongoing series we delve into the unique advantages and innovations that make iron-air batteries a standout choice for sustainable energy storage

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Positive air electrode

For a typical bifunctional Co-Ni oxide air electrode in KOH electrolyte, the predicted equilibrium potential for oxygen is about 0.3V versus Hg/HgO. The actual electrode potentials at which the reduction and the evolution of oxygen occur (at current density ± 20 mA cm^{-2}) are -0.22 and 0.66V vs Hg/HgO, respectively. The overpotential values are 0.52V for reduction and 0.36V for evolution. These large deviations from the reversible value pose a problem in finding a catalyst that can decrease the overpotential.

To compensate for the sluggish rate of oxygen reduction, two air-breathing electrodes are used per iron electrode. To increase efficiency, a fan may be used to supply air, and a pump can be employed in the electrolyte circuit to remove heat through a heat exchanger and for electrolyte purification.

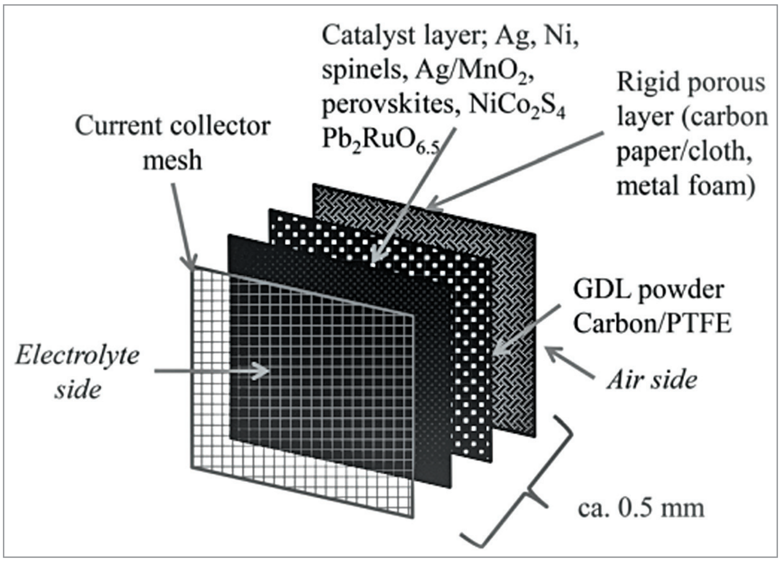

Fig. 10: Exploded view of positive air electrode

It has been identified that MnO2 and Ni-Fe layered double hydroxides are good inexpensive bifunctional catalysts. Additionally, perovskites like La0.5Sr0.4Co0.2Fe0.8O3 (LSCF), LaSr-3Fe3O10 (LSFO), and La0.6Ca0.4CoO3 (LCCO) supported on glassy carbon are alternative air electrode materials that can be used instead of costly conventional noble metal Pt/Pd/Ru-based catalysts.

The reduction of oxygen on perovskites proceeds through a 4e- process, which is more efficient than the 2e- reduction mechanism. The structure of a perovskite also enhances the rate of oxygen evolution. Bifunctional oxygen catalysts have limited electrical conductivity, so a supporting material like carbon blacks is necessary with which the catalyst is bonded, as they combine high surface area with reasonably high conductivity.

As shown in Fig. 10, the air electrode has a more complex design than the iron electrode, as it requires a hydrophilic layer containing the catalyst on the side facing the electrolyte. It also requires a porous hydrophobic layer, termed the 'gas diffusion layer (GDL),' on the reverse side to allow oxygen to diffuse into the electrode.

It is necessary that GDL pores are not flooded with the alkaline electrolyte; hence it is made of microporous conductive carbon powder treated with PTFE (polytetrafluoroethylene) or a similar alkali-resistant hydrophobic polymer. Oxygen from the air diffuses through these open pores towards the hydrophilic catalyst